

CB 4.15 Let $X \sim \text{Poisson}(\theta)$, $Y \sim \text{Poisson}(\lambda)$, independent. It was shown in Theorem 4.3.2 that the distribution of $X + Y$ is $\text{Poisson}(\theta + \lambda)$. Show that the distribution of $(X|X + Y)$ is binomial with success probability $\theta/(\theta + \lambda)$. What is the distribution of $(Y|X + Y)$.

We will find the conditional distribution of $Y|X + Y$. The derivation of the conditional distribution of $X|X + Y$ is similar. Let $U = X + Y$ and $V = Y$. In Example 4.3.1, we found the joint pmf of (U, V) . Note that for fixed u , $f(u, v)$ is positive for $v = 0, \dots, u$. Therefore the conditional pmf is

$$f(v|u) = \frac{f(u, v)}{f(u)} = \frac{\frac{\theta^{u-v} e^{-\theta}}{(u-v)!} \frac{\lambda^v e^{-\lambda}}{v!}}{\frac{(\theta + \lambda)^u e^{-(\theta + \lambda)}}{u!}} = \binom{u}{v} \left(\frac{\lambda}{\theta + \lambda} \right)^v \left(\frac{\theta}{\theta + \lambda} \right)^{u-v}, \quad v = 0, \dots, u.$$

4.17 a. $P(Y = i + 1) = \int_i^{i+1} e^{-x} dx = e^{-i}(1 - e^{-1})$, which is geometric with $p = 1 - e^{-1}$.

b. Since $Y \geq 5$ if and only if $X \geq 4$,

$$P(X - 4 \leq x | Y \geq 5) = P(X - 4 \leq x | X \geq 4) = P(X \leq x) = e^{-x},$$

since the exponential distribution is memoryless.

4.19 (a) $(X_1 - X_2) \sim N(0, 2)$ By example 4.3.4

so

$$\frac{X_1 - X_2}{\sqrt{2}} \sim N(0, 1)$$

$$\left(\frac{X_1 - X_2}{\sqrt{2}} \right)^2 \sim \chi_1^2 \quad \text{By example 2.1.9}$$

(b) $y_1 = \frac{x_1}{x_1 + x_2} \quad y_2 = x_1 + x_2 \Rightarrow \begin{aligned} x_1 &= y_1 y_2 \\ x_2 &= y_2 - y_1 y_2 = y_2(1 - y_1) \end{aligned}$

$$|J| = \begin{vmatrix} \frac{dx_1}{dy_1} & \frac{dx_2}{dy_1} \\ \frac{dx_1}{dy_2} & \frac{dx_2}{dy_2} \end{vmatrix} = \begin{vmatrix} y_2 & -y_2 \\ y_1 & 1 - y_1 \end{vmatrix} = y_2(1 - y_1) + y_1 y_2 = y_2 - y_2 y_1 + y_2 y_1 = y_2$$

$$\begin{aligned} f_{x_1, x_2}(x_1, x_2) &= \frac{1}{\Gamma(\alpha_1)} x_1^{\alpha_1 - 1} e^{-x_1} \frac{1}{\Gamma(\alpha_2)} x_2^{\alpha_2 - 1} e^{-x_2} \\ &= \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)} x_1^{\alpha_1 - 1} x_2^{\alpha_2 - 1} e^{-x_1 - x_2} \end{aligned} \quad \begin{aligned} x_1 &> 0 \\ x_2 &> 0 \end{aligned}$$

Support of y_1, y_2

$y_1 y_2 > 0 \Rightarrow y_1 > 0, y_2 > 0$

$y_2 - y_1 y_2 > 0$

$y_2 > y_1 y_2$

$1 > y_1$

$$f_{y_1, y_2}(y_1, y_2) = \frac{y_2}{\Gamma(\alpha_1)\Gamma(\alpha_2)} (y_1 y_2)^{\alpha_1 - 1} (y_2(1 - y_1))^{\alpha_2 - 1} e^{-y_1 y_2 - y_2 + y_2 y_1} \quad \begin{aligned} y_2 &> 0 \\ 0 < y_1 < 1 \end{aligned}$$

$$= \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_1^{\alpha_1 - 1} (1 - y_1)^{\alpha_2 - 1} y_2^{\alpha_1 - 1} y_2^{\alpha_2 - 1} y_2 e^{-y_2} = \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_1^{\alpha_1 - 1} (1 - y_1)^{\alpha_2 - 1} y_2^{\alpha_1 + \alpha_2 - 1} e^{-y_2}$$

$$= \underbrace{\left[\frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_1^{\alpha_1 - 1} (1 - y_1)^{\alpha_2 - 1} \right]}_{y_1 \sim \text{Beta}(\alpha_1, \alpha_2)} \underbrace{\left[\frac{1}{\Gamma(\alpha_1 + \alpha_2)} y_2^{\alpha_1 + \alpha_2 - 1} e^{-y_2} \right]}_{y_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, 1)}$$

$$y_1 \sim \text{Beta}(\alpha_1, \alpha_2)$$

$$y_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, 1)$$

2. CB 4.20 X_1 and X_2 are independent $\mathcal{N}(0, \sigma^2)$ random variables.

(a) Find the joint distribution of Y_1 and Y_2 , where

$$Y_1 = X_1^2 + X_2^2 \quad \text{and} \quad Y_2 = \frac{X_1}{\sqrt{Y_1}}.$$

This transformation is not one-to-one because you cannot determine the sign of X_2 from Y_1 and Y_2 . So partition the support of (X_1, X_2) into $\mathcal{A}_0 = \{-\infty < x_1 < \infty, x_2 = 0\}$, $\mathcal{A}_1 = \{-\infty < x_1 < \infty, x_2 > 0\}$ and $\mathcal{A}_2 = \{-\infty < x_1 < \infty, x_2 < 0\}$. The support of (Y_1, Y_2) is $\mathcal{B} = \{0 < y_1 < \infty, -1 < y_2 < 1\}$. The inverse transformation from $\mathcal{B}$ to $\mathcal{A}_1$ is $x_1 = y_2\sqrt{y_1}$ and $x_2 = \sqrt{y_1 - y_1 y_2^2}$ with Jacobian

$$J_1 = \left| \begin{array}{cc} \frac{1}{2} \frac{y_2}{\sqrt{y_1}} & \sqrt{y_1} \\ \frac{1}{2} \frac{\sqrt{1-y_2^2}}{\sqrt{y_1}} & \frac{y_2 \sqrt{y_1}}{\sqrt{1-y_2^2}} \end{array} \right| = \frac{1}{2\sqrt{1-y_2^2}}$$

The inverse transformation from $\mathcal{B}$ to $\mathcal{A}_2$ is $x_1 = y_2\sqrt{y_1}$ and $x_2 = -\sqrt{y_1 - y_1 y_2^2}$ with $J_2 = -J_1$. From (4.3.6), $f_{Y_1, Y_2}(y_1, y_2)$ is the sum of two terms, both of which are the same in this case. Then

$$\begin{aligned} f_{Y_1, Y_2}(y_1, y_2) &= 2 \left[\frac{1}{2\pi\sigma^2} e^{-y_1/(2\sigma^2)} \frac{1}{2\sqrt{1-y_2^2}} \right] \\ &= \frac{1}{2\pi\sigma^2} e^{-y_1/(2\sigma^2)} \frac{1}{\sqrt{1-y_2^2}}, \quad 0 < y_1 < \infty, -1 < y_2 < 1. \end{aligned}$$

(b) Show that Y_1 and Y_2 are independent, and interpret this result geometrically.

We see in the above expression that the joint pdf factors into a function of y_1 and a function of y_2 . So Y_1 and Y_2 are independent. Y_1 is the square of the distance from (X_1, X_2) to the origin. Y_2 is the cosine of the angle between the positive x_1 -axis and the line from (X_1, X_2) to the origin. So independence says the distance from the origin is independent of the orientation (as measured by the angle).

3. CB 4.30 Suppose the distribution of Y , conditional on $X = x$ is $\mathcal{N}(x, x^2)$ and that the marginal distribution of X is Uniform(0,1).

(a) Find $E(Y)$, $\text{Var}(Y)$, and $\text{Cov}(X, Y)$

$$\begin{aligned} E(Y) &= E(E(Y|X)) = E(X) = 1/2 \\ \text{Var}(Y) &= \text{Var}(E(Y|X)) + E(\text{Var}(Y|X)) = \text{Var}(X) + E(X^2) = 1/12 + 1/3 = 5/12 \\ E(XY) &= E(E(XY|X)) = E(XE(Y|X)) = E(X^2) = 1/3 \\ \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) = 1/3 - (1/2)^2 = 1/12 \end{aligned}$$

(b) Prove that Y/X and X are independent

Take the transformation $U = Y/X$ and $V = X$. It is not hard to show that $f(u, v) = \sqrt{2\pi}^{-1/2} e^{-u^2/2}$ for $-\infty < u < \infty$ (and $0 \leq v \leq 1$) which doesn't depend of v at all.

4.45 a. We will compute the marginal of X . The calculation for Y is similar. Start with

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right\}\right]$$

and compute

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(\omega^2 - 2\rho\omega z + z^2)\sigma_Y dz},$$

where we make the substitution $z = \frac{y-\mu_Y}{\sigma_Y}$, $dy = \sigma_Y dz$, $\omega = \frac{x-\mu_X}{\sigma_X}$. Now the part of the exponent involving ω^2 can be removed from the integral, and we complete the square in z to get

$$\begin{aligned} f_X(x) &= \frac{e^{-\frac{\omega^2}{2(1-\rho^2)}}}{2\pi\sigma_X\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2(1-\rho^2)}[(z^2 - 2\rho\omega z + \rho^2\omega^2) - \rho^2\omega^2]} dz \\ &= \frac{e^{-\omega^2/2(1-\rho^2)} e^{\rho^2\omega^2/2(1-\rho^2)}}{2\pi\sigma_X\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2(1-\rho^2)}(z-\rho\omega)^2} dz. \end{aligned}$$

The integrand is the kernel of normal pdf with $\sigma^2 = (1-\rho^2)$, and $\mu = \rho\omega$, so it integrates to $\sqrt{2\pi}\sqrt{1-\rho^2}$. Also note that $e^{-\omega^2/2(1-\rho^2)} e^{\rho^2\omega^2/2(1-\rho^2)} = e^{-\omega^2/2}$. Thus,

$$f_X(x) = \frac{e^{-\omega^2/2}}{2\pi\sigma_X\sqrt{1-\rho^2}} \sqrt{2\pi}\sqrt{1-\rho^2} = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2},$$

the pdf of $n(\mu_X, \sigma_X^2)$.

b.

$$\begin{aligned}
& f_{Y|X}(y|x) \\
&= \frac{\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2-2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)+\left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]}}{\frac{1}{\sqrt{2\pi}\sigma_X}e^{-\frac{1}{2\sigma_X^2}(x-\mu_X)^2}} \\
&= \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2-(1-\rho^2)\left(\frac{x-\mu_X}{\sigma_X}\right)^2-2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)+\left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]} \\
&= \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2(1-\rho^2)}\left[\rho^2\left(\frac{x-\mu_X}{\sigma_X}\right)^2-2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)+\left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]} \\
&= \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}}e^{-\frac{1}{2\sigma_Y^2\sqrt{(1-\rho^2)}}\left[(y-\mu_Y)-\left(\rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)\right)\right]^2},
\end{aligned}$$

which is the pdf of $n\left((\mu_Y - \rho(\sigma_Y/\sigma_X)(x - \mu_X), \sigma_Y\sqrt{1 - \rho^2})\right)$.

c. The mean is easy to check,

$$E(aX + bY) = aEX + bEY = a\mu_X + b\mu_Y,$$

Solutions Manual for Statistical Inference

as is the variance,

$$\text{Var}(aX + bY) = a^2\text{Var}X + b^2\text{Var}Y + 2ab\text{Cov}(X, Y) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y.$$

To show that $aX + bY$ is normal we have to do a bivariate transform. One possibility is $U = aX + bY$, $V = Y$, then get $f_{U,V}(u, v)$ and show that $f_U(u)$ is normal. We will do this in the standard case. Make the indicated transformation and write $x = \frac{1}{a}(u - bv)$, $y = v$ and obtain

$$|J| = \begin{vmatrix} 1/a & -b/a \\ 0 & 1 \end{vmatrix} = \frac{1}{a}.$$

Then

$$f_{UV}(u, v) = \frac{1}{2\pi a\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{1}{a}(u-bv) \right)^2 - 2\frac{\rho}{a}(u-bv)v + v^2 \right]}.$$

Now factor the exponent to get a square in u . The result is

$$-\frac{1}{2(1-\rho^2)} \left[\frac{b^2 + 2\rho ab + a^2}{a^2} \right] \left[\frac{u^2}{b^2 + 2\rho ab + a^2} - 2 \left(\frac{b + a\rho}{b^2 + 2\rho ab + a^2} \right) uv + v^2 \right].$$

Note that this is joint bivariate normal form since $\mu_U = \mu_V = 0$, $\sigma_v^2 = 1$, $\sigma_u^2 = a^2 + b^2 + 2ab\rho$ and

$$\rho^* = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{E(aXY + bY^2)}{\sigma_U \sigma_V} = \frac{a\rho + b}{\sqrt{a^2 + b^2 + 2ab\rho}},$$

thus

$$(1 - \rho^{*2}) = 1 - \frac{a^2\rho^2 + ab\rho + b^2}{a^2 + b^2 + 2ab\rho} = \frac{(1-\rho^2)a^2}{a^2 + b^2 + 2ab\rho} = \frac{(1-\rho^2)a^2}{\sigma_u^2}$$

where $a\sqrt{1-\rho^2} = \sigma_U\sqrt{1-\rho^{*2}}$. We can then write

$$f_{UV}(u, v) = \frac{1}{2\pi\sigma_U\sigma_V\sqrt{1-\rho^{*2}}} \exp \left[-\frac{1}{2\sqrt{1-\rho^{*2}}} \left(\frac{u^2}{\sigma_U^2} - 2\rho \frac{uv}{\sigma_U\sigma_V} + \frac{v^2}{\sigma_V^2} \right) \right],$$

which is in the exact form of a bivariate normal distribution. Thus, by part a), U is normal.

4.52

$$\begin{aligned}
X, Y &\stackrel{\text{iid}}{\sim} N(0, 1) \\
(X_1, Y_1) &\sim \\
(X_2, Y_2) &\sim \\
D &= \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2} \\
&\sim X_1 - X_2, Y_1 - Y_2 \sim N(0, 2) \\
&\sim \frac{X_1 - X_2}{\sqrt{2}}, \frac{Y_1 - Y_2}{\sqrt{2}} \sim N(0, 1) \\
D &= \left(\frac{X_1 - X_2}{\sqrt{2}} \right)^2 + \left(\frac{Y_1 - Y_2}{\sqrt{2}} \right)^2 \sim \chi^2_2 \\
&= 2\chi^2_1 + 2\chi^2_1 \\
&= 2(\chi^2_2) \quad \chi^2_2 \sim \text{Gamma}\left(\frac{1}{2}, 2\right) \\
\chi^2_2 &\equiv \boxed{\text{Gamma}(1, 2)} \\
2\chi^2_2 &\sim \text{Gamma}(1, 4) \\
M_X(t) &= (1 - \frac{1}{2}t)^{-2}
\end{aligned}$$

$$\begin{aligned}
M_{2X}(t) &= M_X(2t) \\
2\chi^2_2 &\sim \text{Gamma}(1, 4) \\
u &= 2\chi^2_2 \sim \text{Gamma}(1, 4) \\
V &= \sqrt{u} \\
f(u) &= \frac{1}{\Gamma(1)4^1} u^{1-1} e^{-u/4} \\
&= \frac{1}{4} e^{-u/4} \quad u > 0 \\
u &= v^2, \quad \left| \frac{du}{dv} \right| = 2v \\
f_v(v) &= \frac{2v}{4} e^{-v^2/4} = \frac{v}{2} e^{-v^2/4}, \quad v > 0 \\
v &\sim \text{Weibull}(\kappa=2, \lambda=2)
\end{aligned}$$

4.53

4.53

$$\begin{aligned}
P(\text{Real Roots}) &= P(B^2 > 4AC) \\
&= P(2 \log B > \log 4 + \log A + \log C) \\
&= P(-2 \log B \leq -\log 4 - \log A - \log C) \\
&= P(-2 \log B \leq -\log 4 + (-\log A - \log C)).
\end{aligned}$$

Let $X = -2 \log B$, $Y = -\log A - \log C$. Then $X \sim \text{exponential}(2)$, $Y \sim \text{gamma}(2, 1)$, independent, and

$$\begin{aligned}
P(\text{Real Roots}) &= P(X < -\log 4 + Y) \\
&= \int_{\log 4}^{\infty} P(X < -\log 4 + y) f_Y(y) dy \\
&= \int_{\log 4}^{\infty} \int_0^{-\log 4 + y} \frac{1}{2} e^{-x/2} dx y e^{-y} dy \\
&= \int_{\log 4}^{\infty} \left(1 - e^{-\frac{1}{2} \log 4} e^{-y/2} \right) y e^{-y} dy.
\end{aligned}$$

Integration-by-parts will show that $\int_a^{\infty} y e^{-y/b} = b(a+b)e^{-a/b}$ and hence

$$P(\text{Real Roots}) = \frac{1}{4}(1 + \log 4) - \frac{1}{24} \left(\frac{2}{3} + \log 4 \right) = .511.$$

$$10.) f_{XY}(x,y) = p(1-p)^{x-1} p(1-p)^{y-1} = p^2(1-p)^{x+y-2} \quad \begin{matrix} x=1,2,\dots \\ y=1,2,\dots \end{matrix}$$

a.) transformation $U = X+Y \quad V = Y \text{ or } X$

inverse $X = U-V \quad Y = V$

support of (u,v) : $V=1,2,\dots$
 $X=U-V=1,2,\dots \Rightarrow U=V+1, V+2, \dots$

pmf of (u,v) :

$$f_{UV}(u,v) = f_{XY}(u-v, v) = p^2(1-p)^{u-v+v-2} = p^2(1-p)^{u-2} \quad \begin{matrix} v=1,2,\dots \\ \text{for } u=v+1, v+2, \dots \end{matrix}$$

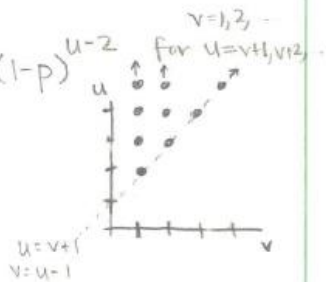
$$f_{UV}(u,v) = p^2(1-p)^{u-2}, \quad \begin{matrix} v=1,2,\dots \\ u=v+1, v+2, \dots \end{matrix}$$

(b) marginal of u

$$f_U(u) = \sum_v f_{UV}(u,v) = \sum_{v=1}^{u-1} p^2(1-p)^{u-2}$$

$$= p^2(1-p)^{u-2} \sum_{v=1}^{u-1} 1$$

$$= (u-1) p^2(1-p)^{u-2} \quad u=2,3,4,\dots$$



This is what we expect for the sum of geometrics:

Note that $U \sim \text{NegBin}(2, p)$

$$\binom{u-1}{r-1} p^r (1-p)^{u-r}$$

$$= \binom{u-1}{1} p^2 (1-p)^{u-2}$$

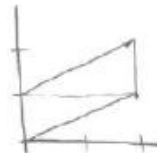
$$= (u-1) p^2 (1-p)^{u-2}$$

10

$$U = 2X \quad V = X+Y$$

$$X = \frac{U}{2} \quad Y = V - \frac{U}{2} \quad |J| = \frac{1}{2}$$

$$f(u,v) = \left[\frac{u}{2} + v - \frac{u}{2} \right] \cdot \frac{1}{2} = \frac{v}{2}, \quad \begin{matrix} 0 < u < 2 \\ \frac{u}{2} < v < 1 + \frac{u}{2} \end{matrix} = f(u,v)$$



$$f(u) = \int_{\frac{1}{2}u}^{1+\frac{1}{2}u} \frac{1}{2} v dv = \frac{1}{4} v^2 \Big|_{\frac{1}{2}u}^{1+\frac{1}{2}u} = \frac{1}{4} \left(1 + \frac{u}{2} \right)^2 - \left(\frac{1}{4} \right) \left(\frac{1}{4} \right) u^2$$

$$f(u) = \frac{1}{4} (1+u) \quad 0 < u < 2$$

$$f(v) = \int_0^{2v} \frac{1}{2} v du = \frac{1}{2} v u \Big|_0^{2v} = \frac{1}{2} v (2v) = v^2 \quad 0 < v < 1$$

$$\int_{2(v-1)}^2 \frac{1}{2} v du = \frac{1}{2} v u \Big|_{2(v-1)}^2 = \frac{1}{2} v (2 - 2v + 2) = v(2-v) \quad 1 < v < 2$$

$$f(v) = \begin{cases} v^2, & 0 < v < 1 \\ v(2-v), & 1 < v < 2 \end{cases} \quad (0 \text{ otherwise})$$

11

Suppose the random variables X and Y have joint density function

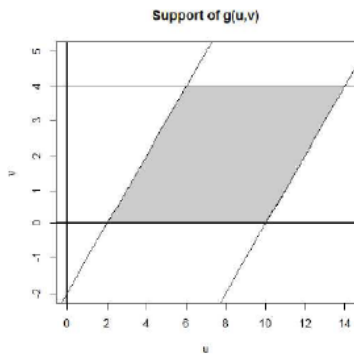
$$f(x, y) = xy/96 \quad 0 < x < 4; 1 < y < 5.$$

Find the density function of $U = X + 2Y$.

Let $V = X$, then the bivariate transformation of $f(x, y)$ to $g(u, v)$ gives

$$g(u, v) = \frac{1}{96} \frac{1}{2} \frac{1}{2} (u - v) = \frac{1}{384} v(u - v) \quad 0 < v < 4 \quad 2 < u - v < 10$$

This support of the joint pdf is shown in the following graph. The marginal pdf for U is the piecewise



function

$$g(u) = \begin{cases} \int_0^{u-2} \frac{1}{384} v(u-v) & 2 < u < 6 \\ \int_0^4 \frac{1}{384} v(u-v) & 6 < u < 10 \\ \int_{u-10}^4 \frac{1}{384} v(u-v) & 10 < u < 14 \end{cases}$$

$$= \begin{cases} \frac{1}{2304} (u-2)^2 (u+4) & 2 < u < 6 \\ \frac{1}{144} (3u-8) & 6 < u < 10 \\ \frac{1}{2304} (348u - u^3 - 2128) & 10 < u < 14 \end{cases}$$

12

$$W = X^2 Y \quad V = X$$

$$Y = \frac{W}{V^2} \quad X = V$$

$$J = \begin{vmatrix} \frac{dx}{dw} & \frac{dx}{dv} \\ \frac{dy}{dw} & \frac{dy}{dv} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ \frac{1}{V^2} & \text{doesn't matter} \end{vmatrix} = \frac{1}{V^2}$$

$$f(x, y) = 2x$$

$$f(w, v) = 2v \cdot \frac{1}{v^2} = \frac{2}{v} \quad \begin{matrix} 0 \leq v \leq 1 \\ 0 \leq w \leq v^2 \end{matrix}$$

$$f(w) = \int_{\sqrt{w}}^1 \frac{2}{v} dv = 2 \log v \Big|_{\sqrt{w}}^1$$

$$= 2 \log(1) - 2 \log(\sqrt{w})$$

$$= -2 \log(\sqrt{w})$$

$$= -\log(w) \quad 0 < w < 1$$